Daniel Arbeláez Álvarez

Software Engineer · University Lecturer · LegalTech Researcher

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PROFILE

Software engineer focused on legal informatics. I build free software for court document management and I teach how to automate court procedures responsibly. I am studying for the Specialization in Artificial Intelligence at UNIR. I am a public servant of the Judicial Branch of Colombia.

Technologies: Python (FastAPI, Streamlit, pandas, spaCy), R, SQL (PostgreSQL, SQLite, SQL Server), Power BI, Power Automate, Docassemble, Visual Basic, Java, Git and GitHub, Docker, Bash.

EDUCATION

Specialization in Artificial Intelligence · Universidad Internacional de La Rioja (UNIR) — In progress, since 2025

Software Engineering · Tecnológico de Antioquia — 2021

Thesis project: web application for monitoring and managing air-quality information in Segovia, Antioquia (institutional repository, in Spanish).

Diplomas

- Soft skills · Universidad Sergio Arboleda (2025).
- Citizen Data Science · Escuela Superior de Administración Pública, ESAP (2024).
- Competency-Based Pedagogy (online) · Politécnico Mayor (2024).
- Computer Crimes, Cyber Risks and Digital Evidence Handling · Universidad Externado de Colombia (2023).

EXPERIENCE

Judicial Branch of Colombia · Public servant — November 2014 – present

Since 2024, technology support and judicial information systems.

Tecnológico de Antioquia · Adjunct lecturer — 2024

Backup Tecnología (Medellín) · Technical support assistant — March 2012 – July 2014

Preventive and corrective maintenance of computers and printers; software installation and updates; support for voice and data networks.

FREE SOFTWARE AND PROJECTS

AgilEx (formerly GestionExpedienteElectronico) — Since 2022 · MIT

Creates and fills in the electronic index of a court case file from the metadata of the files in a folder (Agreement PCSJA20-11567 of 2020). First commit: 5 January 2022. DNDA copyright registration: 3 November 2022 (version 1). Repository.

TYBABot — 2025 · MIT

Conversational assistant that guides users on Justicia XXI Web (TYBA). Built from a template by another author. Repository.

DisciplinaJudicialAI — 2026 · MIT

Conversational assistant that guides judicial servants on the disciplinary regime, without advice on specific cases. Built from a template by another author. Repository.

Docassemble Expert — 2025 · MIT

Conversational assistant for writing and debugging Docassemble YAML interviews. Built from a template by another author. Repository.

TutelaBot (adaptation) — 2025 · MIT (original author)

Adaptation of an assistant on the acción de tutela and the right of petition, published under the MIT license by another author. Repository.

CDAID — Since 2026 · in development

Methodology and repository template for developing software with AI agents from specifications. Code not published.

Other public repositories: statistical analysis of experimental data in R (2024); chat interface for the Claude and GPT APIs (2025); data extraction with the Spotify API (2022); data analysis with Power BI (2022); visualization of PM10 particulate-matter data in Python (2020). github.com/HammerDev99.

RECOGNITION AND ACADEMIC ACTIVITY

- **2024.** Member of team Marduk, winner of the 1st JusticIALab Innovation Contest, Data-Driven Justice category (Superior Council of the Judiciary news, 13 December 2024, in Spanish).
- **2024.** Shared talk "National case studies: Colombia's experiences incorporating AI into the justice system", in which I presented the Marduk case. Academic Conference on Artificial Intelligence and Its Impact on Justice, Medellín, 18 October 2024 (programme, in Spanish).
- **2024.** Arbeláez Álvarez, D. Entrevistas guiadas para la optimización de procesos judiciales: un enfoque desde el LegalTech [Guided interviews for optimizing court procedures: a LegalTech approach]. *Cuaderno Activa*, 16(1), 37–47. doi:10.53995/20278101.1858.
- **2015.** Student representative on the Faculty of Engineering Council, Tecnológico de Antioquia.
- **2013–2014.** Speaker at the 12th and 13th Regional Meetings of Research Seedbeds (Corporación Universitaria Lasallista, 2013; Politécnico Colombiano Jaime Isaza Cadavid, 2014) and at the Tecnológico de Antioquia Research Days (2013).

CERTIFICATIONS AND COURSES

2025 · MinTIC · Artificial Intelligence Bootcamp, intermediate level

2025 · ESAP · Introduction to Public Innovation

2025 · Platzi · Machine Learning with Python

2024 · Rodrigo Lara Bonilla Judicial School · ICT and Its Use in Litigation

2024 · Tecnológico de Antioquia · Cybercrime and Its Legal Concepts

2024 · Función Pública · Integrity, Transparency and Anti-Corruption

2023 · Universidad de los Andes · AI Project Formulation for the Justice Sector

2023 · Intelligent Training · Task Automation with Power Automate

2023 · Rodrigo Lara Bonilla Judicial School · Training for Systems Engineers of the Judicial Branch

2023 · Coderhouse · Cybersecurity

2023 · Platzi · Python Comprehensions, Functions and Error Handling

2022 · Platzi · Python Fundamentals

2022 · Platzi · Python: PIP and Virtual Environments

2022 · Tecnológico de Antioquia · Information Security Fundamentals

2022 · Rodrigo Lara Bonilla Judicial School · Training in Information and Communication Technologies

2022 · UiPath · RPA Business Analysis Fundamentals

2022 · UiPath · Introduction to RPA and Automation

2022 · Platzi · Object-Oriented Programming

2022 · Platzi · English Course for Beginners

2022 · Platzi · Business Analysis for Data Science

2022 · Platzi · Data Science and Artificial Intelligence

2022 · Platzi · Python Basics

2022 · Coderhouse · Data Analytics

2022 · Platzi · OOP and Algorithms in Python

2022 · Platzi · Computational Thinking with Python

2020 · Coursera · Python Data Visualization

2020 · Coursera · Neural Networks and Deep Learning

Certificates: public folder.

Personal opinions; I do not represent the Judicial Branch. I do not comment on cases or give advice. Teaching, research and free software.